Utkrishta 2026 — AI/ML Hackathon

Official Participant Guidelines & Problem Statements

Welcome Participants: This document contains official problem statements, constraints, evaluation metrics, and submission requirements for Utkrishta 2026. Select your challenge track and adhere strictly to all evaluation rules.

Challenge 1 — Autonomous AI Data Scientist

Theme: Budget-Constrained Experiment Optimization for Churn Prediction

Problem Objective

Build an AI/data-science system that predicts customer churn while operating under a strict budget of model-training experiments. The key challenge is not simply achieving the highest predictive score through brute-force trial-and-error; your system must intelligently decide which experiments are worth running using feedback from previous runs.

Track Specifications & Evaluation Rules

Component / Metric	Specification & Participant Rules
Dataset	**Telco Customer Churn Dataset** (Single-table tabular data provided in your participant portal).
Strict Constraint	**Maximum 15 Model-Training Experiments.** One fit-and-validation cycle counts as 1 experiment run. Data loading, cleaning, profiling, and feature design do NOT count.
Primary Metric	**ROC-AUC Score** evaluated on organizer-held hidden test labels.
Secondary Metric	**PR-AUC / Average Precision** to evaluate ranking on imbalanced customer churn.
Leaderboard Formula	Primary predictive score scaled by budget compliance: `Final Score = ROC-AUC × Efficiency Factor`

Permitted Approaches

- **LLM-Assisted Planners:** Utilizing LLM prompts to analyze prior run logs and recommend optimal hyperparameter configurations.
- **Optimization & Search:** Bayesian optimization, bandit algorithms, or heuristic search strategies.
- **Standard Classifiers:** Logistic Regression, Random Forests, XGBoost, LightGBM, CatBoost, etc.

Required PPT Content for Challenge 1

- Problem formulation, feature engineering, and data preprocessing decisions.
- Experiment planner strategy and a complete chronological log/table of all 15 executed runs.
- Validation performance trajectory across iterations and final hidden test score.

Challenge 2 — Sentinel

Theme: Adaptive Time-Series Anomaly Detection

Problem Objective

Build a lightweight anomaly-detection system that learns baseline behavior from streaming time-series data and identifies observations or windows that deviate significantly. For every flagged event, your system must produce anomaly scores, a binary NORMAL/ANOMALOUS label, and interpretable evidence explaining the flag.

Track Specifications & Metrics

Metric / Component	Purpose & Description
Dataset	**Numenta Anomaly Benchmark (NAB)** subset prepared in the starter pack with hidden test windows.
Primary Metric	**Event-Based F1 Score** (measures detection accuracy across temporal anomaly windows).
Precision & Recall	Evaluates true positive anomaly flags against total flagged windows and true ground-truth events.
Detection Delay	Measures latency between true anomaly onset and model trigger timestamp.
Inference Runtime	Measures practical execution speed per streaming data window.

Permitted Approaches

- **Statistical Baselines:** Rolling z-scores, Isolation Forests, Local Outlier Factor (LOF), and One-Class SVMs.
- **Advanced Models:** Autoencoders, temporal clustering, sliding-window adaptive thresholds, and hybrid ensembles.

Required PPT Content for Challenge 2

- Time-series feature representation, windowing strategy, and adaptive threshold logic.
- Detailed precision, recall, event-F1, and detection latency performance metrics.
- Visual examples of correctly flagged anomaly windows, false positive analysis, and evidence explanations.

Participant Deliverables

1. **Source Code & Notebooks:** Fully functional code repository with instructions to run locally.
2. **README Documentation:** Detailed setup, environment requirements, experiment strategy, and reproducibility steps.
3. **Prediction Results File:** Standardized outputs formatted according to organizer submission schemas.
4. **Final Slide Deck (PPT):** 10-slide presentation matching the required structure below.
5. **Live / Video Demo:** Brief working demonstration of your pipeline or application.

Required 10-Slide Deck Structure

Slide	Required Content Guidelines
Slide 1	Team name, team members, chosen track, and one-line solution summary.
Slide 2	Problem understanding, practical motivation, and real-world significance.
Slide 3	Dataset breakdown, exploratory data analysis, and preprocessing workflow.
Slide 4	System design diagram and end-to-end processing pipeline.
Slide 5	Core algorithm design, model choices, or automated planner logic.
Slide 6	Experiment history log, tuning trajectory, and hyperparameter search strategy.
Slide 7	Evaluation methodology, validation strategy, and metric breakdowns.
Slide 8	Results summary tables, comparison with baseline models, and charts.
Slide 9	Demo screenshots, sample predictions, or anomaly evidence outputs.
Slide 10	Known failure modes, system limitations, future work, and conclusion.

Evaluation & Judging Breakdown

Criteria	Weight	What Judges Will Evaluate
Technical Performance	**40%**	Primary score on organizer hidden evaluation data.
Approach & Innovation	**20%**	Algorithmic intelligence, creative engineering, and threshold design.
Efficiency & Constraints	**15%**	Strict compliance with 15-run limit (Ch 1) or runtime speed (Ch 2).
Presentation Quality	**15%**	Clarity, structure, visualization quality, and Q&A defense.
Code & Demo Quality	**10%**	Code organization, documentation completeness, and working demo.